

FLATWORMS

Structurally among the simplest of animals, flatworms live wherever a moist habitat can provide their thin, flat body with oxygen and food.

Platyhelminthes, commonly known as flatworms, are a major phylum of animals found in various forms in the ocean, in freshwater ponds, or even within the bodies of other animals. Superficially resembling leeches, flatworms are in fact far simpler animals. With no blood system and no organs for breathing, they use their entire body surface to absorb oxygen and release carbon dioxide. Flatworm species that have no gut also absorb their food in this way. In other species the gut has one opening but branches frequently, so that digested food can still reach all tissues,

even without a blood stream to circulate the nutrients. Some free-living flatworms are detritus-eaters, gliding along on a bed of microscopic beating hairs. Others prey on other invertebrates.

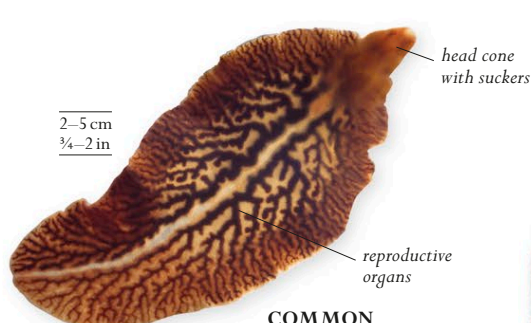
INTERNAL PARASITES

Tapeworms and flukes are parasites. Inside the host animal, their flattened body is perfectly suited for absorbing nutrients. Many have complex ways of passing from host to host and may infect more than one type of animal. They enter the body of the host in infected food or sometimes by penetrating the skin. Once inside, a parasite may go deeper into the body – burrowing through the gut wall to lodge in vital organs.

PHYLUM	PLATYHELMINTHES
CLASSES	6
ORDERS	41
FAMILIES	About 420
SPECIES	About 30,000

DEBATE A NEW PHYLUM?

Traditionally, the tiny, gutless, and brainless marine animals called acoels are viewed as flatworms. Controversial studies suggest that they belong in a new phylum of their own, and that they are the most primitive living animals to show bilateral symmetry – in contrast to radial (round) cnidarians.



COMMON LIVER FLUKE
Fasciola hepatica
F: Fasciolidae

Flukes typically have complex, parasitic life cycles. This species infects grass-eating sheep or cattle via freshwater snails, and feeds from the host's liver.



JACK FLUKE
Provitellus turrum
F: Monorchidae

Provitellus flukes infect the guts of fish living on tropical reefs. This species, shown here in false colour, has been recorded in jack fish.



CATTLE-SNORING BLOOD FLUKE
Schistosoma nasale
F: Schistosomatidae

This parasite is transmitted through freshwater snails. It infects cattle, causing nasal growths that make the cattle snore when breathing.



EPIZOIC FLATWORM
Waminoa sp.
F: Convolutidae

This coral-living species is one of many marine micro-flatworms that resemble planktonic larvae of cnidarians.



GIANT INTESTINAL FLUKE
Fasciolopsis buski
F: Fasciolidae

This E. Asian species is one of the largest flukes to infect humans. It lives only in the upper intestine and does not enter other organs.



GIANT LEAF FLATWORM
Kaburakia excelsa
F: Stylochidae

Like other leaf flatworms, this N. American intertidal species is mainly carnivorous. It smothers its prey with a mouth-bearing extension of its gut.



PORK TAPEWORM
Taenia solium
F: Taeniidae

Tapeworms are intestinal parasites, but at an intermediate stage can invade muscle as cysts. This species' intermediate host is pigs, and may pass to humans in infected meat.



CANDY STRIPE FLATWORM
Prostheceraeus vittatus
F: Euryleptidae

This animal belongs to a group of mainly marine, free-living, cross-fertilizing hermaphrodites called polyclads. It is an Atlantic Ocean species.